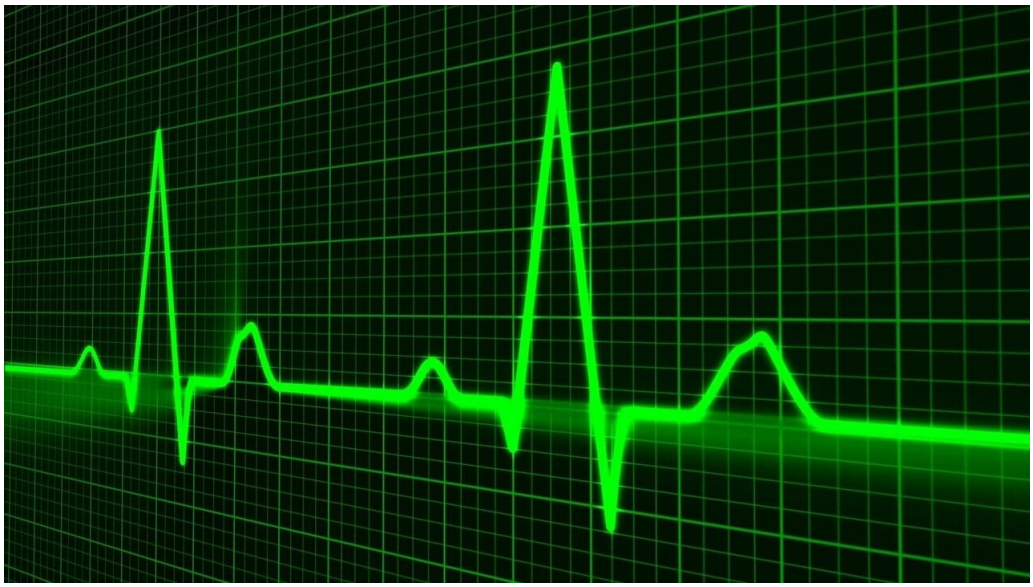
LIÈGE université
Sciences Appliquées

GBIO0029 - Bioelectronics

ECG Front-end



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1 Introduction

The goal of this project is to design an ECG front-end in order to physically measure the heart-rate of a student.

1.1 ECG signal decomposition

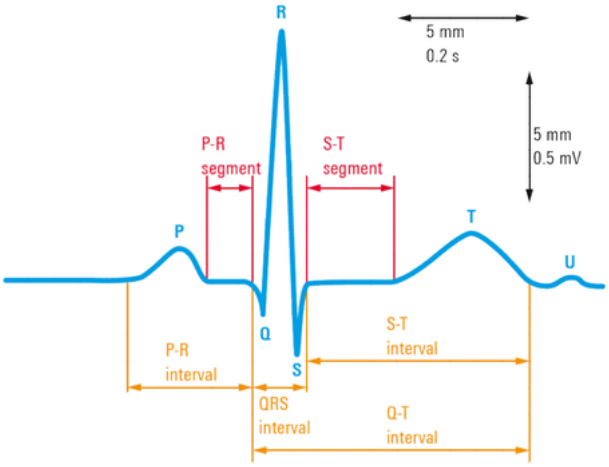


Figure 1: Decomposition of the different phases of an ECG signal

The graph in Figure 1 represents a standard ECG waveform, illustrating the electrical activity of the heart beat over time. It is composed of these four main stages:

- **P Wave**: Represents atrial depolarization, indicating the activation of the atria.
- **QRS Complex**: Corresponds to ventricular depolarization. The sharp spikes represent:
  - **Q Wave**: Initial negative deflection.
  - **R Wave**: Prominent positive deflection.
  - **S Wave**: Following negative deflection.
- **T Wave**: Represents ventricular repolarization, showing the recovery of the ventricles.
- **U Wave**: Occasionally present, possibly related to Purkinje fiber repolarization. Not a lot of focus will be placed on this wave.

There are also 5 different time intervals labelled. They can provide insights to the cardiac function and help determine if there are anomalies or problems in a patient.

The typical amplitude of an ECG wave is 0.5-1.5mV and the duration of the wave is around 400-600ms. In order to measure these signals, 3 electrodes will be used. They will be respectively placed on the left arm, right arm and right leg.

1.2 Challenges

The main design challenges are the low voltage generated by a heartbeat, which is in the order of a few mV. There are also many common-mode parasitics and interferences, for example: capacitive coupling with the body or power line interference. Another challenge is the imbalance between the 3 different electrodes.

In order to address these problems, several systems were put in place to obtain the cleanest possible ECG signal at the output. These different parts are detailed below in the rest of the report.

2 Circuit description and components

2.1 Circuit description

The complete circuit of the ECG front-end modelled in *LT Spice* can be seen in Figure 2. It is based off the example shown in the chapter 5 of the course : "ECG front-end". Several modifications and differences were implemented in comparison to the course example. These helped achieve a better signal in this specific real life implementation.

The left of the schematic houses the equivalent circuit for the electrodes and the right leg driven amplifier. Whilst the right of the circuit has all the signal processing components such as the instrumentation amplifier, notch filter or several RC filters. The utility and implementation of each specific stage is detailed below. They will be analysed by progressing from left to right in the schematic.

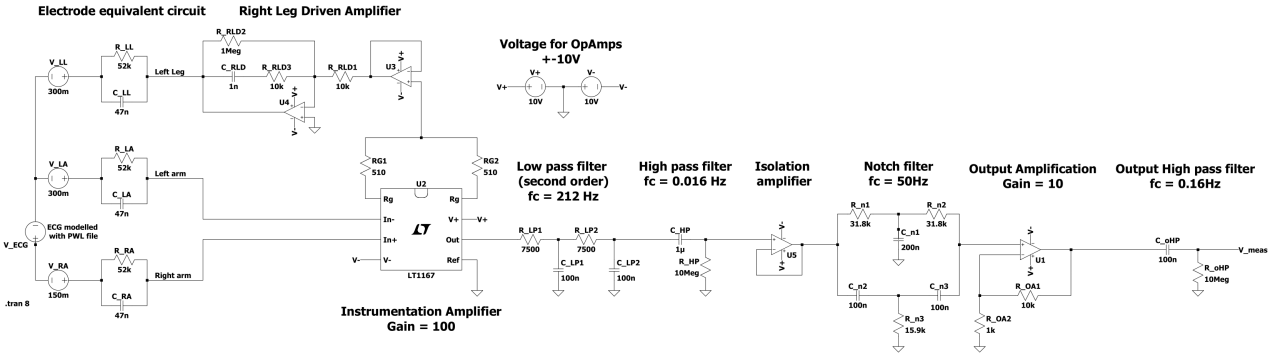


Figure 2: Complete circuit of the ECG front-end modelled in *LT Spice*

2.2 Electrodes

Figure 3 shows the equivalent diagram used to simulate the electrodes in *LT Spice*. This layout is based on the theoretical model from the course. However, real-world electrodes are not perfectly ideal or aligned with regards to one another. Therefore certain non-idealities were introduced. As shown in the diagram the voltage source, capacitor and resistance values vary slightly to account for these imperfections.

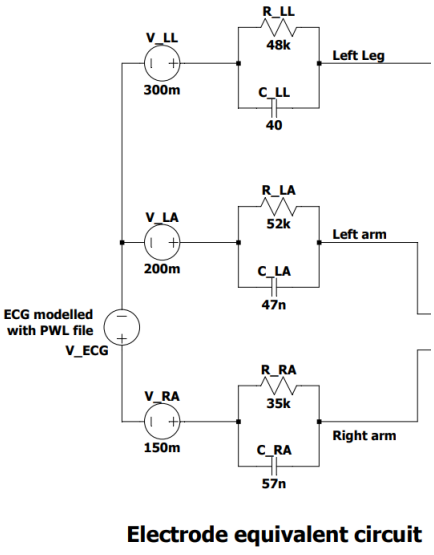


Figure 3: Equivalent diagram for the electrodes

2.3 Right leg driven amplifier

The right leg driven amplifier is a piece of the circuit used to reduce as much as possible the common mode noise on the output of the ECG. The RLD amplifier drives the common mode voltage through a negative feedback path to the body, this sets a common reference point.

The output is connected to the right leg of the patient and the input is taken between the two resistors $R_g/2$ of the instrumentation amplifier (their value is value 210Ω), this input is the common mode voltage of the signals coming from the two arms electrodes.

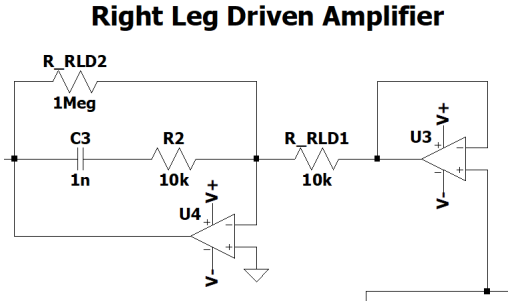


Figure 4: Circuit diagram for the right led driven amplifier

As can be seen on Figure 4, the first stage is a unity gain buffer, it is used to make a galvanic isolation between the main circuit and the second stage of the RLD amplifier. This stage copies its input voltage at the output, without any power transfer between the input and the output.

The second stage then is used to manipulate the common mode voltage at the input in order to set a reference voltage to the body that is as stable as possible. The more stable the reference voltage, the better the reduction of the common mode noise. This is why the values of the components of the circuit had to be wisely selected to ensure stability of the system with the feedback path provided by the RLD amplifier. Firstly, the higher the gain of the second stage, the better the reduction of the common mode, according to the computation made during the theoretical lectures. The expression of the voltage gain of the second stage at low frequencies is :

$$G = \frac{R_{RLD2}}{R_{RLD1}} = \frac{10^6}{10^4} = 100V/V$$

This meant that the common mode voltage is reduced by a factor of 101 (it is not perfect but sufficient, the reason why this value was chosen is explained below) because the expression of v_{cm} is

$$v_{cm} = i_d * \frac{Z_{el}}{1 + \frac{R_{RLD2}}{R_{RLD1}}}$$

Here, i_d is the current through the right leg electrode and Z_{el} is the equivalent impedance of the electrode. Normally, having only this path should be enough to reduce the common mode noise. However, stability issues arise due to the pole of the inverting amplifier in the RLD. The frequency of this pole depends on the values of R_{RLD1} and R_{RLD2}, and more precisely on the gain, as can be seen on Figure 5. In our case, we thus have a pole at 10 kHz.

R_2 is chosen equal to R_{RLD1} to have a high feedback factor at high frequencies. Then, the value of the capacity is chosen to have a zero close to the pole. We chose 1 nF, that makes a zero frequency of :

$$f_z = \frac{1}{2\pi C_3 R_2} = \frac{1}{2\pi 10^{-9} 10^4} = 15,915kHz$$

Then, the first reason why the value of 100 for the gain was chosen is to have a pole frequency above the expected common mode interference frequencies (50 Hz). It is important to have that pole above the interference to keep a good rejection of the common mode noise. In this case, the pole frequency is :

$$f_p = \frac{1}{2\pi C_3 R_{RLD2}} = \frac{1}{2\pi 10^{-9} 10^6} = 159,15Hz$$

Some tests were done to verify that our RLD amplifier worked correctly by disconnecting it at this point. We also tried to change some values to see if it could work better, but the results were not very effective. Even with the changes, the common mode voltage was still oscillating the same way. We supposed that the reduction was the highest possible and to reduce more the noise, other external parameters as the place or the addition of a shield had to be considered.

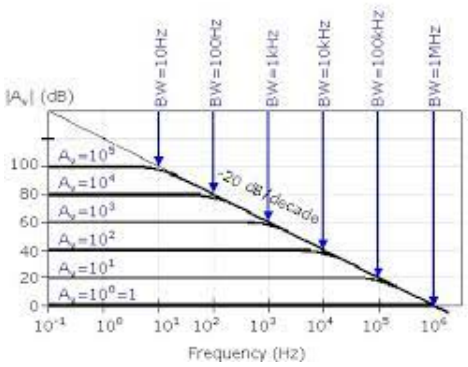


Figure 5: Frequency response of an inverting amplifier

2.4 Instrumentation amplifier

The instrumentation amplifier used in this application is an INA128 from Texas Instruments. This component is used to amplify the differential signal coming from the electrodes whilst getting rid of noise.

As mentioned in the introduction, the signal generated by the body is only in the order of a few mV. Therefore significant amplification is required to obtain a useful signal. The amplifier has a very high input impedance so not to affect the small input voltage. The difference between the two inputs $V+$ and $V-$ is the signal that is amplified. The gain for this amplification is calculated by the formula from the data-sheet which can be seen on the left of Figure 7 and is:

$$A = 1 + \frac{50k\Omega}{R_g} = 1 + \frac{50k\Omega}{120 \cdot 2} \approx 418 \qquad \text{(rounded to 420)}$$

The second important part of this amplifier is to get rid of the common mode noise caused by the power supply and other sources of noise. This component has a very high common mode rejection ratio. This is done with the help of the 2 first amplifiers and the third input coming from the RLD amplifier (see section 2.3). Measuring differential inputs naturally removes common mode noise between the two inputs.

The use of the RLD helps provide a stable reference voltage from the body. Grounding can be an issue for a human body, therefore this acts as a virtual ground for the system. This is therefore used for the two internal OpAmps (A1 and A2 in Figure 7) The third OpAmp (A3) is then used to further amplify the signal before being sent to the output.

It is important to mention that the resistance values of R_g displayed in Figure 6 represent the values placed in the real life circuit (120Ω). For the simulation however, resistances of 510Ω were used. At high gain, the instrumentation amplifier model displays peculiar behaviour and saturates, no longer letting a suitable signal reach the output. This could be due to the oversimplification of the INA128 in the equivalent *LT Spice* model or maybe stray capacitance and impedances could unintentionally stabilise the circuit in real life(but this is a far reach). The two options to solve this issue was to increase R_g or increase the supply voltage to the amplifier. But as this supply voltage is identical for all the OpAmps, the first solution of increasing R_g was implemented. Therefore, the simulation works fine with 510Ω resistors for R_g, but the simulated gain is slightly lower than the one in real life (100 instead of 500).

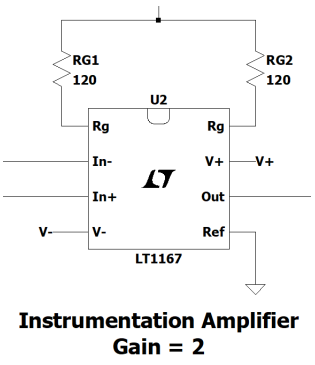


Figure 6: Circuit diagram for the instrumentation amplifier

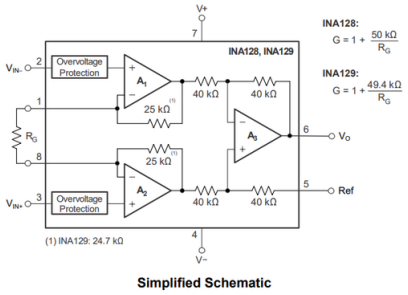


Figure 7: Equivalent diagram of the internal structure of the instrumentation amplifier [1]

[1] Instrumentation amplifier INA 128 datasheet

2.5 RC filters

An ECG signal comprises frequencies ranging from 0.05 Hz to 150 Hz. To filter out components outside this range, we have chosen a second-order low-pass filter implemented by cascading two first-order filters, in series with a first-order high-pass filter.

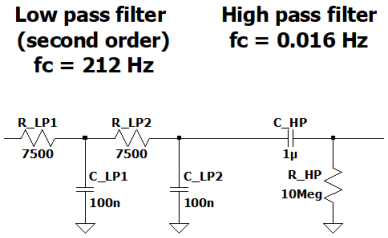


Figure 8: Second order low pass filter and first order high pass filter placed before the notch filter

For the low-pass filter, we have a cutoff frequency of 212.02 Hz. We opted to cascade two first-order filters to achieve a second-order response, which accelerates the attenuation of noise related to other bioelectric signals and electromagnetic interference. This approach provides a steeper roll-off beyond the cutoff frequency compared to a single first-order filter, which is beneficial for suppressing high-frequency noise. By selecting appropriate resistor R and capacitor C values, we ensure the overall cutoff frequency aligns with our design requirements.

The high-pass filter's role is to remove the DC components of the signal (the offset), which can result from electrode potentials or other sources. Its cutoff frequency is set at 0.016 Hz, which is low enough to preserve the ECG signal components starting from 0.05 Hz while effectively eliminating the DC offset. The high-pass filter is implemented as a first-order filter, which is sufficient for attenuating the DC component without significantly affecting the low-frequency components of the ECG signal. The cutoff frequency f_c is calculated using:

$$f_c = \frac{1}{2\pi R_{\text{HP}} C_{\text{HP}}} = 0.016 Hz$$

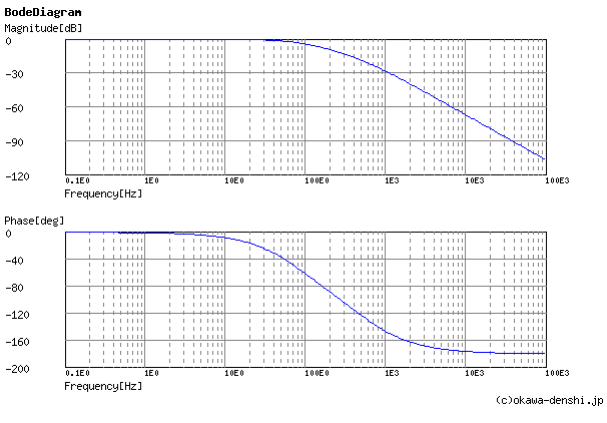


Figure 9: Bode plot of the second order low pass filter

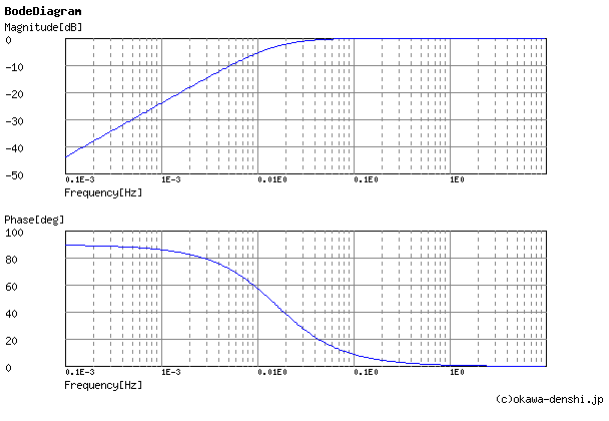


Figure 10: Bode plot of the high pass filter

- **Low-Pass Filter Components:**
  - Resistors: $R_{LP1} = R_{LP2} = 7.5k\Omega$
  - Capacitors: $C_{LP1} = C_{LP2} = 100nF$

- **High-Pass Filter Components:**
  - Resistor: $R_{HP} = 10M\Omega$
  - Capacitor: $C_{HP} = 1\mu F$

2.6 Isolation amplifier

An isolation amplifier was placed before the notch filter. The configuration of placing it after the notch was also examined but the chosen setup has proven to be the most effective at keeping good signal stability.

The main goal of the isolation amplifier is to provide electrical isolation between the ECG electrodes and the rest of the circuit. This is useful for patient safety by preventing any currents from returning to the patient. Additionally, the isolation amplifier breaks ground loops, which makes the circuit more stable to noise.

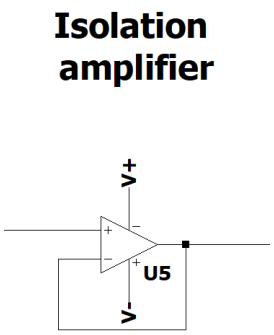
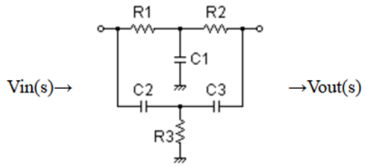


Figure 11: Circuit diagram for the isolation amplifier before the notch filter

2.7 Notch filter

A vital frequency component that must be attenuated is the 50Hz component of the mains power. As this is a very specific frequency which is very stable, a notch filter can be used. The goal of this filter is to only remove the 50Hz without affecting the rest of the signal. A simple bandstop, passive layout was used in this case. The value of the different components are detailed in Figure 12.



Transfer function:

$$G(s)=\frac{s^3+314.46540880503s^2+98888.493334916s+31097010.482678}{s^3+1572.3270440252s^2+494442.46667458s+31097010.482678}$$

Center rejection frequency

$f_0 = 50.048724242734$[Hz]
$f_0 = 50.048724242734$[Hz]

R1=	31.8k	Ω	C1=	200n	F
R2=	31.8k	Ω	C2=	100n	F
R3=	15.9k	Ω	C3=	100n	F

Figure 12: Diagram and cut-off frequency calculation [2]

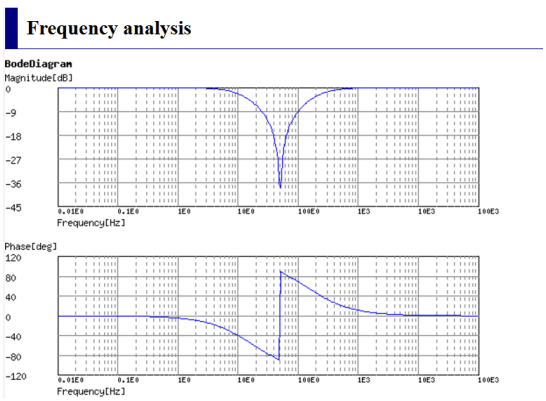


Figure 13: Bode plots for the notch filter

[2] Okawa Electric design website used for the cut off frequency calculation

2.8 Output amplification phase

This final output stage is used to amplify the signal with a gain of 10. This helps obtain a higher voltage signal at the output. This enables easier detection and readings on the oscilloscope. The gain is calculated as :

$$A = 1 + \frac{R_{OA1}}{R_{OA2}} = \frac{10k\Omega}{1k\Omega} = 10$$

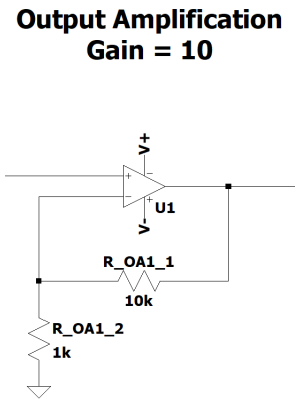


Figure 14: Circuit diagram for the output amplification stage

2.9 Output high pass filter

The final stage of the circuit is the output high pass filter shown in Figure 15. The goal of this final filter is to get rid of 2 things: any potential DC offset and low frequency variations due to the patients movement.

The final amplification stage may introduce a DC offset. If the op-amps are not ideal, they can generate unwanted DC signals that need to be removed.

Small movements made by the patient like respiration for instance can cause variations in the measurement. These would be of very low frequency. Therefore the choice of a cut-off frequency of 0.015Hz allows to get rid of these perturbations.This frequency is calculated as follows:

$$f_c = \frac{1}{2\pi R_{\text{oHP}} C_{\text{oHP}}} = 0.15Hz$$

This has proven to help with the overall stability of the signal in the real life tests.

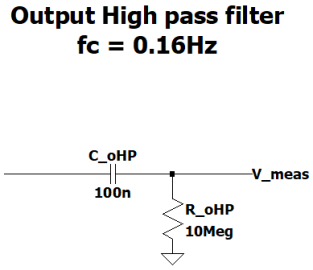


Figure 15: Circuit diagram for the output high pass filter

3 Results

3.1 LTspice

After replicating each component of the circuit in *LT Spice*, it is possible to simulate the entire ECG signal. To more closely approximate reality, a noisy input signal was used during the simulations, and non-idealities were represented by varying some of the electrode components. Figures 16 and 17 show the input and output potentials. As expected, high-frequency noise is attenuated, and the offset is filtered by the low-pass filters. The effect of gain can also be observed on the ECG signal, increasing from 3.6 mV to 1.4 V peak-to-peak.



Figure 16: Noisy input signal voltage

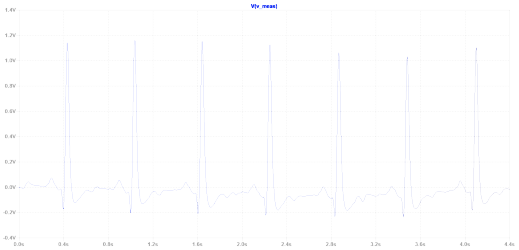


Figure 17: Amplified output signal voltage

3.2 In practice

Figure 18 shows, despite some remaining noise, a satisfying output signal from the physical circuit. Despite attempts to increase the gain of the Right Leg Driven (RLD) amplifier, the results remained very similar. The characteristics of an healthy ECG signal can easily be identified, such as the QRS complex corresponding to ventricular depolarization (and contraction) and the T wave corresponding to ventricular repolarization (and relaxation). In some cases, even the smaller waves, such as the P wave and the U wave, were recognizable.



Figure 18: ECG signal visualized on the oscilloscope

4 Conclusion

4.1 Problems encountered

4.1.1 Delay of observed signal

Initially, only a capacitor was added at the end of the circuit, which led to some issues. When connected to the oscilloscope, the capacitor discharged through the oscilloscope's input impedance, which is relatively low. However, upon disconnection of the oscilloscope, the capacitor remained charged due to the absence of a discharge path. This resulted in delays when reconnecting the oscilloscope, as time was required for the capacitor to discharge through the oscilloscope before the signal could be properly observed.

To address this issue, a resistor was added in parallel with the capacitor, forming an output high-pass filter. This configuration provided a continuous discharge path for the capacitor, even in the absence of the oscilloscope, thereby ensuring smoother signal observation.

4.1.2 Reduction of common-mode noise

At the end of the completion of the circuit, a significant amount of time was spent attempting to reduce the impact of common-mode noise. The following approaches were tested:

- **Internal modification** : The gain of the RLD was increased. This was intended to improve the ability of the RLD to reject common-mode signals by amplifying the reference signal, making the difference between the desired signal and the common-mode noise more distinct. However, the increase in gain also amplified other noise sources, which could lead to a less effective overall noise reduction.
- **External modification** : The circuit was placed in a metallic box, which is similar to the principle of a Faraday cage, as it helps block or reduce electromagnetic interference.
- **External modification** : The cables connecting the electrodes from the arms to the circuit were twisted, which was intended to cancel out some of the external electromagnetic interference.

Unfortunately, none of these attempts to reduce common-mode noise were successful. No hypothetical or theoretical explanations were found for the failure of these techniques. Further investigation and adjustments may be needed to better address the source of the failure.

4.2 Future improvements

Here is a list of additional improvements that could be made:

- The arrangement of components within the circuit could be further optimized. Prolonged cable lengths can introduce signal degradation due to increased parasitic capacitance and inductance, which may impact signal integrity, especially in high-precision applications such as ECG measurement.
- Redesigning the circuit on PCB could enhance the overall quality of the system by improving signal integrity, reducing electromagnetic interference, and mitigating the effects of common-mode noise.
- It was observed that the prolonged or repetitive use of the same electrodes led to a noticeable decrease in the measured signal over time. It could be due to the degradation of electrode contact resistance, which affects signal acquisition. The use of higher-quality electrodes or other type of electrodes, could improve the accuracy and reliability of heart rate measurements over time.

A Appendix

A.1 Photos of the physical circuit

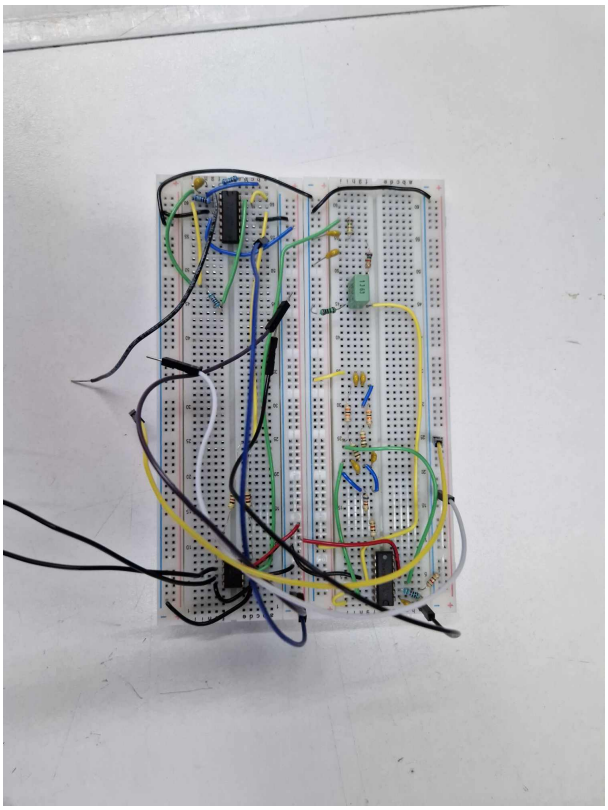


Figure 19



Figure 20

A.2 Photo of the test subject



Figure 21: The happy test subject